7

Order and Randomness

Where Probability Comes From

One sunny day in Cambridge, England, Elizabeth Anscombe ran into her teacher, Ludwig Wittgenstein. "Why do people say," Wittgenstein opened in his inimitable fashion, "that it was natural to think that the sun went round the earth, rather than that the earth turned on its axis?" Anscombe gave the obvious answer, that it just *looks like* the sun goes around the Earth. "Well," Wittgenstein replied, "what would it have *looked like* if the Earth had turned on its axis?"

This anecdote—recounted by Anscombe herself, and which Tom Stoppard retold in his play *Jumpers*—is a favorite among Everettians. Physicist Sidney Coleman used to relate it in lectures, and philosopher of physics David Wallace used it to open his book *The Emergent Multiverse*. It even bears a family resemblance to Hugh Everett's remark to Bryce DeWitt.

It's easy to see why the observation is so relevant. Any reasonable person, when first told about the Many-Worlds picture, has an immediate, visceral objection: it just doesn't *feel like* I personally split into multiple people whenever a quantum measurement is performed. And it certainly doesn't *look like* there are all sorts of other universes existing parallel to the one I find myself in.

Well, the Everettian replies, channeling Wittgenstein: What would it feel and look like if Many-Worlds were true?

The hope is that people living in an Everettian universe would experience just what people actually do experience: a physical world that seems to obey the rules of textbook quantum mechanics to a high degree of